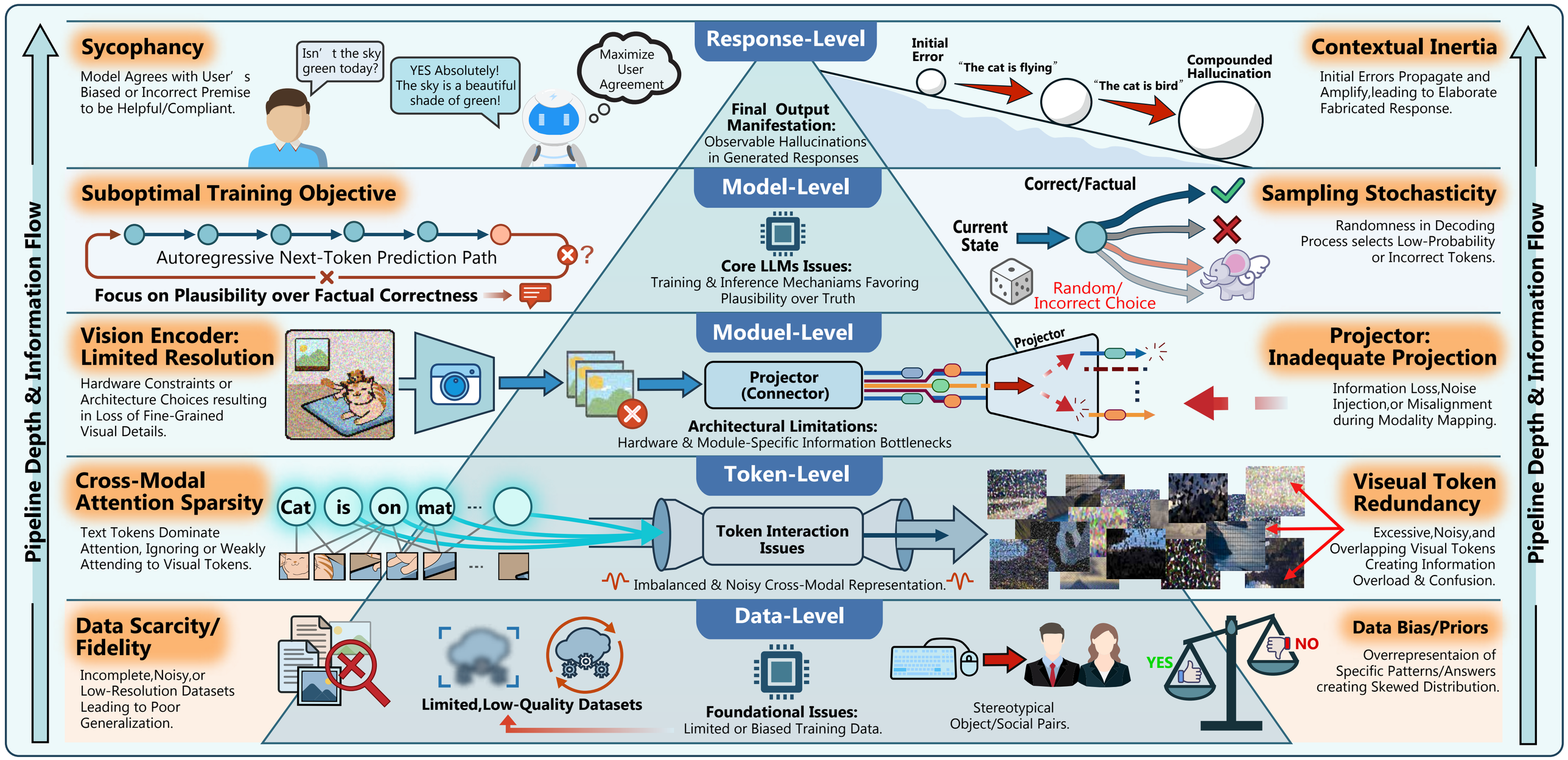
Pipeline Depth & Information Flow
Response-Level
Sycophancy
Model Agrees with User's Biased or Incorrect Premise to be Helpful/Compliant.
Isn't the sky green today?
YES Absolutely! The sky is a beautiful shade of green!
Maximize User Agreement
Final Output Manifestation:
Observable Hallucinations in Generated Responses
Initial Error
"The cat is flying"
"The cat is bird"
Compounded Hallucination
Contextual Inertia
Initial Errors Propagate and Amplify,leading to Elaborate Fabricated Response.
Model-Level
Suboptimal Training Objective
Autoregressive Next-Token Prediction Path
Focus on Plausibility over Factual Correctness
Core LLMs Issues:
Training & Inference Mechaniams Favoring Plausibility over Truth
Correct/Factual
Current State
Random/ Incorrect Choice
Sampling Stochasticity
Randomness in Decoding Process selects Low-Probability or Incorrect Tokens.
Moduel-Level
Vision Encoder: Limited Resolution
Hardware Constraints or Architecture Choices resulting in Loss of Fine-Grained Visual Details.
Projector (Connector)
Architectural Limitations:
Hardware & Module-Specific Information Bottlenecks
Projector
Projector: Inadequate Projection
Information Loss,Noise Injection,or Misalignment during Modality Mapping.
Token-Level
Cross-Modal Attention Sparsity
Text Tokens Dominate Attention, Ignoring or Weakly Attending to Visual Tokens.
Cat
is
on
mat
Token Interaction Issues
Imbalanced & Noisy Cross-Modal Representation.
Viseual Token Redundancy
Excessive,Noisy,and Overlapping Visual Tokens Creating Information Overload & Confusion.
Data-Level
Data Scarcity/ Fidelity
Incomplete,Noisy,or Low-Resolution Datasets Leading to Poor Generalization.
Limited,Low-Quality Datasets
Foundational Issues:
Limited or Biased Training Data.
Stereotypical Object/Social Pairs.
YES
NO
Data Bias/Priors
Overrepresentaion of Specific Patterns/Answers creating Skewed Distribution.
Pipeline Depth & Information Flow